

Composition Pair B1.18 + B1.19 — Content-Domain Boundary Specifications and Cross-Level Access Governance as a Reinforcing Pair: Operational Territory Defines What Access Reaches; Access Governance Extends Boundary Enforcement Across Levels

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to establish, in operational form, that the B1.18 content-domain boundary commitment and the B1.19 cross-level access governance commitment are a reinforcing pair — each enabling properties in the other that neither produces alone — and to name the two single-commitment failure modes that emerge when one is present without the other.

Abstract

The Coordination Knowledge Substrate (CKS) pattern extended by Paper 2 commits to a three-level architecture — cell, aspect, Self — in which higher levels operate over lower levels as content domain, and in which cross-level access patterns are not strictly hierarchical (Paper 2, §5.2). Two architectural commitments govern this structure: the content-domain boundary specification (B1.18), which authors the operational territory each entity covers and the rules enforcing that territory; and cross-level access governance (B1.19), which specifies which entities at which levels can interact with which entities at other levels. This note formalizes these two commitments as a reinforcing pair. B1.18 requires B1.19 when cross-level interactions affect content-domain boundaries: without a governed cross-level access framework, inputs arriving from a different structural level may bypass the boundary enforcement machinery that operates within an entity’s own scope. B1.19 requires B1.18 for access scope definition: without content-domain specifications, access rules can govern who reaches which entities but cannot specify what operational territory that access reaches, making access technically governed but semantically hollow. Their mutual reinforcement produces governance that is both authority-based (who can access) and semantically specific (what operational territory is being accessed): the double governance that complete inter-level interaction governance requires. Two single-commitment failure modes name the deficits of partial instantiation: B3.19 (Implicit Content-Domain) and B3.20

(Ungoverned Cross-Level Access).

1. Why this pair requires formalization

Paper 2's three-level architecture (§5.2) creates a structural circumstance that neither commitment addresses alone. Higher levels operate over lower levels as content domain: an aspect coordinates cells and asks questions across those cells for its purpose; a Self holds multiple aspects as facets of one integrated intelligence. But access patterns in this architecture are not strictly hierarchical — the Self can access cells directly when purpose requires, aspects can operate on cells from above, and expression mechanisms (B1.07) move activation from Self-level governance down to cell-level substrates. Cross-level access is not an edge case; it is a structural property of how the three levels relate.

This creates two governance questions that a single commitment cannot answer simultaneously. The first question is *what* is being reached when cross-level access occurs: what operational territory does the accessed entity cover, and what boundary constraints govern what it handles? The second question is *who* can reach it and how: which entities at which levels are governed to interact with which entities at other levels, and through what access pathway?

A system that answers only the first question has authored content-domain specifications for each entity but no governed pathway through which cross-level interactions can be held to those specifications. Boundary enforcement operates within an entity's own scope; inputs arriving from outside that scope through an ungoverned cross-level channel enter without passing through the enforcement machinery. A system that answers only the second question has access rules that specify traversal patterns across levels but no semantic grounding in what each accessed entity handles. The access rules know who can reach which addresses; they do not know what operational territory each address covers or what boundary constraints should govern the interaction.

The reinforcing pair supplies both answers together, and the co-presence is what makes cross-level interaction governance complete.

2. The two commitments

B1.18 — Content-domain boundary specification. In Paper 2's three-level architecture, each entity — cell, aspect, or Self — covers a specific operational territory: the informational tasks a cell handles, the purpose-defined coordination a given aspect manages, the integrated scope a Self holds. The B1.18 commitment requires that this operational territory be *authored* as substrate content rather than left implicit. Authored content-domain specifications take the form of rules, schemas, and scope definitions that constitute the entity's boundary specification — the explicit statement of what the entity handles and, by exclusion, what falls outside its territory. Boundary enforcement per B2.91 operates when inputs arrive: the entity's authored boundary specification provides the reference against which arriving inputs are evaluated, with violations recorded as first-class events per Paper 1's conflict-preservation commitment (§5) and surfaced as A1.03-class conflicts when relevant.

The content-domain commitment is an authored property, not an inferred one. An entity whose operational territory is implied by the context of its use rather than explicitly specified in substrate content does not satisfy B1.18. The specification must be substrate content — human-governed, inspectable, and modifiable — for the governance properties Paper 1 established (§3.1) to extend to the entity’s scope definition itself.

B1.19 — Cross-level access governance. The B1.19 commitment requires that inter-level interactions — any case where an entity at one structural level interacts with an entity at another structural level — occur through governed access pathways rather than ungoverned channels. Governed access per B2.95 takes the form of access rules specifying which entities at which levels can interact with which entities at other levels, under what conditions, and through what mechanisms. These access rules are themselves substrate content, authored under human governance. The Self’s direct access to cells (§5.2) is a governed pattern, not unconstrained traversal; expression mechanisms (B1.07) that activate cell-level substrates from Self-level governance are access patterns governed by the rules carrying the expression specification.

Cross-level access governance does not collapse the three levels into one; it specifies how the levels relate to each other operationally. The governance character of cross-level access — the fact that access rules are substrate content under human authority — is what makes it an extension of Paper 1’s governance architecture (§3.3) across the full three-level structure.

3. B1.18 requires B1.19 when cross-level interactions affect content-domain boundaries

Content-domain boundary enforcement per B2.91 operates within an entity’s own scope: when an entity receives inputs through its own direct interface, its authored boundary specification is the reference against which those inputs are evaluated. This is the within-level case, and it is fully covered by B1.18 alone.

The cross-level case is structurally different. When a cell receives inputs through an expression mechanism (B1.07) — a Self-level governance pattern that activates cell-level substrates — those inputs arrive through a cross-level channel. When an aspect coordinates cells for its purpose by querying their content, those queries arrive at the cell from the aspect level. In both cases, the input source is an entity at a different structural level, and the path from that entity to the cell passes through inter-level interaction machinery rather than through the cell’s own direct input interface.

Without B1.19’s governed cross-level access framework, that inter-level interaction machinery has no architectural obligation to incorporate the cell’s content-domain boundary specifications. The expression mechanism activates substrates without reference to which substrates are within the cell’s content-domain specification; the aspect-level query reaches cells without reference to what those cells are authored to handle. The boundary enforcement machinery that B1.18 provides operates within the entity’s scope — but the inputs have already arrived by the time the within-scope machinery engages, and the cross-level channel through which they traveled had no mechanism for screening them.

B1.19's governed access framework is the structural location where this gap closes. When cross-level access rules per B2.95 are authored with reference to content-domain specifications, those rules can incorporate boundary constraints into the access pathway itself: the access rule governing which entities at which levels can interact with which other entities can specify that access is authorized only within the target entity's content-domain scope. The B2.97 cross-level access verification checks whether access rules incorporate content-domain-informed scope as a requirement of access authorization.

The B1.18 commitment thus requires B1.19 whenever the deployment includes cross-level interactions — which, in Paper 2's three-level architecture (§5.2), is architecturally pervasive. An architecture that authors content-domain specifications without also governing cross-level access has boundary enforcement that operates locally but cannot follow the entity through its participation in cross-level interaction patterns.

4. B1.19 requires B1.18 for access scope definition

Cross-level access rules per B2.95 specify traversal patterns: which entities at which levels can reach which entities at which other levels. This specification is structurally complete — it names the source-entity level, the target-entity level, and the conditions of authorized access. What it does not specify, without B1.18, is what the accessing entity reaches *as operational territory* when it accesses the target.

The structural address of an entity — cell C at aspect A within Self S — tells a cross-level access rule where to route. It does not tell the access rule what operational territory cell C covers, what informational tasks it is authored to handle, or what boundary constraints govern what should enter it. Without B1.18's content-domain specifications, cross-level access rules are technically governed but semantically unspecific. An access rule that says “Self can access cell C” is complete as an authority specification; it is silent as an architectural specification of what the Self reaches by accessing cell C.

This semantic gap matters because the purpose of cross-level access in Paper 2's architecture is not traversal for its own sake. When the Self accesses a cell directly (§5.2), it does so because the cell's operational territory is relevant to the Self's integration scope. When an expression mechanism activates cell-level substrates, it does so because the substrates addressed fall within the cell's authored content-domain for the current cell goal. The purpose of the access is to reach a specific operational territory; the operational territory is what makes the access meaningful rather than arbitrary.

B1.18 supplies the missing semantics. When cross-level access rules per B2.95 reference entities whose content-domain specifications per B1.18 are authored substrate content, the access rules become architecturally specific in two ways. First, access authorization can incorporate content-domain alignment: “the Self is authorized to access cells whose content-domain addresses the Self's integration scope for this query” is a rule with semantic content, not merely a structural traversal specification. Second, the access event has an interpretable purpose: accessing cell C through a governed access pathway means reaching an entity with content-domain Y, where Y is authored substrate content that the accessing entity's orchestration rules can reference. The access is purposeful rather than positional.

The B1.19 commitment thus requires B1.18 to fulfill the semantic dimension of access governance. An architecture that authors cross-level access rules without content-domain specifications governs the mechanism of inter-level interaction but not its meaning.

5. Mutual reinforcement: what the pair produces

The two enabling directions above establish dependency. Mutual reinforcement is the additional claim: that the co-presence of B1.18 and B1.19 produces a property — semantically grounded cross-level boundary governance — that neither commitment can produce alone even when both their individual contributions are fully realized.

B1.18 provides semantic depth to B1.19 access rules. When cross-level access rules per B2.95 reference entities with authored content-domain specifications per B1.18, the access rules gain semantic specificity. Access governance becomes a two-dimensional commitment: the authority dimension (who can access which entities across which levels) and the semantic dimension (what operational territory is being accessed in each authorized case). The composition compatibility check per B2.92 can then evaluate both dimensions simultaneously — not only whether two entities' access patterns are compatible structurally, but whether the content-domain specifications of the entities involved are compatible with the access relationships the rules authorize.

B1.19 provides governance enforcement across levels for B1.18 boundaries. Content-domain boundary enforcement per B2.91 is a within-scope mechanism; it catches violations when they arrive at the entity through pathways the entity's own boundary machinery governs. B1.19's governed access architecture extends this enforcement to cross-level pathways: when access rules per B2.95 incorporate content-domain boundary constraints, the enforcement begins at the access pathway rather than only at the entity's own boundary. This earlier enforcement point — at the cross-level access gateway rather than at the entity's boundary — is what complete cross-level boundary governance requires.

The pair's combined governance is: inter-level interactions are governed both by who can access (B1.19) and by what the accessed entity handles (B1.18). The double governance is not redundant. Each dimension catches what the other misses: authority-based governance without semantic grounding permits access to entities for purposes outside their content-domain; semantically-grounded content-domain enforcement without access governance has no pathway-level mechanism to apply those semantics cross-level.

6. Co-presence analysis and single-commitment failure modes

B1.18 without B1.19 — B3.19: Implicit Content-Domain. When content-domain specifications are absent, cross-level access rules exist and govern inter-level traversal, but the access is technically governed without semantic specificity. The access rules specify authority patterns — which entities at which levels can reach which other entities — but cannot specify what operational territory is being accessed in each authorized case. Downstream effects propagate into the deployment's behavior: aspects access cells without reference to what those cells are authored to handle; expression mechanisms activate substrates without a specification of content-domain alignment; composition

compatibility checks per B2.92 operate on structural patterns but have no semantic dimension to evaluate. The system governs the mechanism of cross-level access; it does not govern whether that access is reaching entities for purposes within their authored operational territory.

B1.19 without B1.18 — B3.20: Ungoverned Cross-Level Access. When cross-level access governance is absent, content-domain specifications are authored and boundary enforcement per B2.91 operates within each entity's scope, but cross-level interactions bypass that enforcement. Inputs arriving through expression mechanisms (B1.07), through aspect-level queries to cells, or through Self-level direct cell access enter entities through pathways that carry no architectural obligation to respect the entity's content-domain specification. Boundary violations per B2.91 that originate from within-level input pathways are caught; violations that originate from cross-level channels are not, because those channels have no governed specification that makes content-domain boundary constraints applicable at the pathway level. The system has authored operational territory specifications; it does not have the architectural machinery to enforce those specifications when the interaction source is at a different structural level.

Full co-presence. When both commitments are present, cross-level access rules per B2.95 reference entities whose content-domain per B1.18 is authored substrate content. Access governance is both authority-based and semantically specific. Content-domain boundary constraints inform the design of cross-level access rules at authoring time; the access rules carry those constraints into the cross-level interaction pathway; boundary enforcement per B2.91 applies at both the pathway level (via access rules) and the entity boundary level (via the entity's own enforcement machinery). A1.03-class conflicts arising from content-domain boundary violations through cross-level access channels are registered as first-class objects per Paper 1's conflict-preservation commitment (§5), with their origin in cross-level access recorded in the provenance metadata per A1.07. Composition compatibility per B2.92 evaluates both content-domain compatibility and cross-level access compatibility together.

7. Operational integration and detection

At deployment design. Entities receive authored content-domain specifications per B1.18 as part of their substrate configuration: the scope definitions, schemas, and boundary rules that specify what each entity handles are authored substrate content, inspectable and modifiable under the human-governance commitment. Cross-level access rules per B2.95 are configured in reference to those content-domain specifications: the access rule design asks which higher-level entities need cross-level access to which lower-level entities and for what content-domain purposes, and the rules' scope conditions incorporate content-domain-informed constraints.

During operation. When cross-level access occurs per B1.19's governed pathways, the accessing entity reaches an entity whose content-domain it can identify from the substrate. Expression mechanisms (B1.07) that activate cell-level substrates do so through access pathways that incorporate the target cell's content-domain specification in the activation scope. Boundary violations arriving through cross-level channels are detectable by the entity's boundary enforcement machinery per B2.91 and are registered as first-class conflict objects with provenance traceable to the cross-level access

event.

Detection criteria for pair completeness. Four checks collectively assess whether the pair is fully instantiated:

1. *B2.91 boundary enforcement check:* Do the entity's boundary enforcement rules address inputs arriving through cross-level access channels, not only inputs arriving through within-level pathways?
2. *B2.95 access rule check:* Do cross-level access rules reference content-domain specifications in their scope conditions, or do they specify only structural traversal authority without semantic grounding?
3. *B2.97 cross-level access verification:* Does verification of cross-level access incorporate content-domain-informed scope confirmation as a component of access authorization, not merely structural access authorization?
4. *Pair completeness check:* When a cross-level access event is traced in the substrate's provenance records per A1.07, does the trace include both the access authority basis (B1.19) and the content-domain scope of the accessed entity (B1.18)?

A deployment that passes all four checks instantiates the reinforcing pair. A deployment that passes the B2.95 check without the B2.91 cross-level extension has B1.19 without full B1.18 enforcement across levels. A deployment that passes the B2.91 boundary check within-scope but fails the B2.95 content-domain scope condition has B1.18 without B1.19's semantic grounding.

8. Conclusion

The B1.18 content-domain boundary specification and the B1.19 cross-level access governance commitment are a reinforcing pair in Paper 2's three-level architecture. Neither commitment is complete without the other when the deployment includes cross-level interactions — and in a three-level architecture where higher levels operate over lower levels as content domain and access patterns are not strictly hierarchical (§5.2), cross-level interactions are architecturally pervasive rather than exceptional.

B1.18 requires B1.19 because the boundary enforcement machinery that content-domain specifications enable operates within an entity's own scope; without a governed cross-level access framework, inputs arriving from other levels enter through pathways that carry no architectural obligation to respect that boundary specification. B1.19 requires B1.18 because cross-level access rules that specify only structural traversal authority — without semantic grounding in what operational territory each accessed entity covers — govern the mechanism of inter-level interaction but not its purpose. The pair's mutual reinforcement produces governance that is simultaneously authority-based and semantically specific: inter-level interactions governed by who can access (B1.19) and by what the accessed entity handles (B1.18).

Two single-commitment failure modes name the respective deficits: B3.19 (Implicit Content-Domain), where access is technically governed but the semantic dimension of what operational territory is being

accessed is absent from the access rules; and B3.20 (Ungoverned Cross-Level Access), where content-domain specifications are authored but their boundary constraints cannot extend across level boundaries because no governed cross-level access framework is present to carry them. The full reinforcing pair forecloses both failure modes simultaneously.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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